

AM335x BeagleBone Black GPIO + I2C Reference

GPIO Bank Structure:

GPIO0 : 0 - 31 (32 pins)

GPIO1 : 32 - 63 (32 pins)

GPIO2 : 64 - 95 (32 pins)

GPIO3 : 96 - 127 (32 pins)

Key Rule:

Total GPIOs = 4 banks x 32 pins = 128

Valid GPIO range = 0 to 127

Formula:

GPIO number = (32 x bank) + pin

Example:

GPIO1_31 = (32 x 1) + 31 = 63

Failure Case:

If GPIO = 200:

200 > 127

Therefore, it is outside the valid range and invalid.

Kernel returns: Invalid argument error.

I2C Configuration:

I2C Slave Address: 0x34

This address is used for I2C communication with a slave device.

Ensure the address is not conflicting with other devices on the same I2C bus.